AI-POWERED SUPPLY CHAIN RISK INTELLIGENCE ASSISTANT

Overview

Supply chain teams need to continuously monitor suppliers, orders, inventory, delivery performance, disruptions, and operational risks. When this information is spread across multiple datasets, identifying potential risks and understanding their business impact can be difficult.

Users may ask questions such as:

- "Which suppliers have a high risk of delivery delays?"
- "Which products are most affected by supplier disruptions?"
- "Identify unusual changes in supplier lead times."
- "Which suppliers should be investigated based on recent performance?"
- "What are the major supply chain risks and what evidence supports them?"
- "How could a supplier delay affect inventory availability?"

Instead of manually analyzing supplier and operational data, the system should allow users to ask natural-language questions and receive data-driven risk insights supported by the underlying supply-chain data.

Build an **AI-Powered Supply Chain Risk Intelligence Assistant** that combines supplier-performance analysis, risk detection, anomaly detection, forecasting, natural-language querying, and AI-generated risk insights to help supply-chain teams identify and investigate potential operational risks.

Tasks

Task	Task Overview	Weightage
Task 1 – Supply Chain Data Preparation & Risk Knowledge Base	Prepare the supply-chain data and build the foundation for intelligent risk analysis.	**20%**
Task 2 – Supplier Performance, Risk Detection & Forecasting	Analyze supply-chain performance and identify potential supplier and operational risks.	**25%**
Task 3 – Multi-Agent Supply Chain Risk Intelligence & Decision Support	Extend the analytics system into a specialized multi-agent workflow for deeper supply-chain risk analysis.	**25%**
Task 4 – Evaluation, Risk Reporting & User Application	Evaluate the risk-intelligence capabilities and provide a simple application for supply-chain users.	**20%**
Task 5 – Architecture, Documentation & Demonstration	Document the complete solution and demonstrate the working Supply Chain Risk Intelligence Assistant.	**10%**
Total		**100%**

Task Details

Task 1 – Supply Chain Data Preparation & Risk Knowledge Base

- Prepare the provided supply-chain dataset for intelligent analysis. The solution should process relevant information such as:
    - Suppliers
    - Products
    - Orders
    - Deliveries
    - Inventory
    - Locations
    - Dates/time periods
- Create appropriate analytical features such as delivery delay, supplier lead time, order frequency and supplier-performance indicators where supported by the dataset.
- Data-quality checks and preprocessing decisions should be documented.

Task 2 – Supplier Performance, Risk Detection & Forecasting

- Build the core supply-chain risk analytics capability. The solution should identify:
    - Supplier performance issues
    - Recurring delivery delays
    - Unusual lead-time changes
    - Abnormal order patterns
    - Potential inventory/availability risks
    - Products or orders affected by supplier risks
- A suitable **risk-scoring approach** should be implemented and clearly documented. Where the dataset supports it, the participant should implement forecasting for demand, inventory, lead time or another relevant supply-chain variable.
- Forecasting should be evaluated using suitable metrics such as **MAE, RMSE or MAPE**.

Task 3 – Multi-Agent Supply Chain Risk Intelligence & Decision Support

- Extend the analytical system into a specialized multi-agent supply-chain risk workflow.
- The implementation should include:
    - **Supplier Risk Analysis Agent** – analyzes supplier performance and identifies supplier-level risk indicators.
    - **Delivery Risk Agent** – analyzes delivery delays, lead times and delivery patterns.
    - **Inventory Risk Agent** – identifies potential stock-out or product-availability risks.
    - **Demand/Forecasting Agent** – analyzes future demand or relevant supply-chain forecasts.
    - **Supply Chain Insight Agent** – consolidates findings from the specialist agents.
    - **Risk Recommendation Agent** – provides data-backed recommendations for investigation or mitigation.

- The system should also demonstrate:
    - Agent handoff for complex risk scenarios.
    - **A2A communication** where applicable.
    - Evidence-backed risk assessments.
    - Traceability between generated insights and the underlying supply-chain data.

  For example, if the system reports:
  "Supplier A has elevated delivery risk."

Task 4 – Evaluation, Risk Reporting & User Application

Evaluate the quality of the risk-intelligence system. The evaluation should consider:

- Supplier-risk identification
- Anomaly-detection effectiveness
- Forecast accuracy where applicable
- Natural-language query accuracy
- Correctness of generated risk insights
- Evidence grounding

The front-end should allow supply-chain users to:

1. View supplier-performance information.
2. Identify potential supplier risks.
3. View delivery or inventory anomalies.
4. View relevant forecasts.
5. Ask natural-language supply-chain questions.
6. Receive risk insights.
7. View supporting data/evidence.

Task 5 – Architecture, Documentation & Demonstration

- Provide complete technical documentation and demonstrate the working solution.
- The architecture should clearly show:
  Supply Chain Data → Data Processing → Analytics → Risk Detection / Forecasting → Multi-Agent Intelligence → Risk Insights → User
- The documentation should explain the major technical decisions, risk-scoring approach, analytical models, agent workflow, evaluation methodology and design trade-offs.

Deliverables

1. Architecture Diagram

- Include a high-level view of your system in JPEG or PDF.
- Show how data is processed from the provided supply-chain dataset through to the user's query and final risk-intelligence response.

2. Design

- Articulate system design decisions taken and the trade-offs made.

3. Full Executable Code (Microservice)

- Focus on clarity and modularity.
- Provide a README explaining:
    - Project setup (how to install or run)
    - Sample usage (e.g., a supply-chain risk query and the resulting analysis/insight)

4. Panel Presentation (10 minutes)

- Demo of your working solution outlining the design and approach (8 minutes).
- QCA (2 minutes).

Dataset Name

Accenture Supply Chain Resilience Challenge Dataset

Primary Link

Accenture Supply Chain Resilience Challenge – Kaggle

Alternative Links

DataCo Global Supply Chain – Kaggle

Supply Chain Logistics Dataset – Kaggle

ProcureSense Supply Chain Dataset – GitHub

Format

CSV

Key Fields

Orders dataset:

- order_id
- origin_port
- 3pl
- customs_procedure
- logistic_hub
- customer
- product_id
- units
- late_order

Product attributes:

- product_id
- weight
- material_handling

City/logistics data:

- Origin and destination cities
- Coordinates
- distance

The primary dataset specifically includes **historical inbound/outbound shipment transactions** and a late_order field indicating whether an order was delivered late.

Reason

The dataset provides order, logistics, product, transportation and delivery information required to demonstrate supply-chain risk identification, late-delivery prediction, logistics analysis, anomaly detection, supplier/3PL risk analysis and AI-generated supply-chain insights. The late_order field provides a concrete target for building and evaluating a delivery-risk model.